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ARTICLE

Inferring the Relative Oceanic Distribution of Salmon from Patterns in Age-Specific Arrival Timing

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Abstract

The oceanic distribution of Columbia River spring–summer Chinook Salmon *Oncorhynchus tshawytscha* is essentially unknown, as they are first detected in the fishery at the mouth of the river and then at Bonneville Dam, located 235 km upstream of the river mouth. Although the composite run timing to Bonneville Dam varies interannually, earlier studies revealed that the distinct populations comprising this run are ordered in arrival time. Our analysis of returning tagged adults shows that within populations, the oldest fish (3-ocean; i.e., those that spend 3 years in the ocean) arrive first, the 2-ocean fish arrive next, and the 1-ocean fish arrive last. To explore possible mechanisms underlying this persistent age-dependent ordering of arrival time, we simulated the fish's return migration using a model based on a geomagnetic homing mechanism with age-specific swimming speeds. To reproduce the observed age-dependent differences in arrival time with the model, the 2-ocean and 3-ocean fish within a population needed to initiate migration from the same location and time, while 1-ocean fish could initiate homeward migration from a more consistent location in the northern Gulf of Alaska with some delay. Arrival timing patterns provide valuable information that can be used to infer the features of homing migrations, which in turn may be useful for managing fish harvest by integrating age information into existing management techniques and for considering potential responses to climate change.

Pacific salmon *Oncorhynchus* spp. spend several months to several years traversing the North Pacific Ocean before beginning their directed migration back to their home river and spawning grounds. It is believed that this distinct stage in the life cycle evolved to return fish to their natal site contemporaneously with others spawning in the same location, thus maintaining local adaptations and a population's genetic integrity (Quinn 2005). However, salmon runs in large river systems typically consist of multiple populations, each with unique, genetically determined arrival timing patterns (Quinn et al. 2000; Waples et al. 2004) that are subject to environmental variability (Keefer et al. 2004). For example, the many distinct populations that constitute the Columbia River spring–summer Chinook Salmon *O. tshawytscha* run exhibit consistent patterns in their arrival timing, with some populations reliably returning early in the migration season and others returning late (Keefer et al. 2004). The relative order of population arrival is preserved from year to year even though the timing of the combined run varies interannually by up to a month (Anderson and Beer 2009).

Some evidence suggests age dependence in arrival timing (Jonsson et al. 1990; Shearer 1990; McKinnell et al. 1994; Newell et al. 2007). Columbia River spring Chinook Salmon jacks (precocious adult males) return later in the season than older adults (Beer 2007; Anderson and Beer 2009), although this is not true of all salmon species and populations (Morbey 2000; Carlson et al. 2004). The persistence of arrival time

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ordering among populations (and possibly among ages) is striking. Spring Chinook Salmon spend their ocean residence mostly offshore, but their spatial distribution in the vast area (~5 million km^2) of the Northeast Pacific Ocean is unknown (Healey 1991). However, it seems that the temporal pattern of the composite run would be unlikely to persist without some degree of spatial–temporal cohesion in the initiation of homeward migration among populations and age-groups within the run.

A better understanding of this important stage of the salmon life cycle is valuable in its own right, but it could also be used to improve harvest management in cases where overlapping arrival patterns of populations make the harvest of mixed-stock runs problematic. Uncertainties in run timing and size make it difficult to distribute harvest equally over all populations and ages while meeting spawner escapement goals; delaying harvest to meet escapement goals can selectively remove late-arriving populations and younger fish, while fixing harvest across the run may selectively remove early returning fish, particularly if the escapement goal is revised in-season (Flynn et al. 2006; Newell et al. 2007; Anderson and Beer 2009). Several models have predicted run timing based on environmental covariates—generally a combination of in-river and oceanic indices (Burgner 1980; Mundy 1982; Blackbourn 1987; Dahl et al. 2004; Hodgson et al. 2006; Keefer et al. 2008; Thomson and Hourston 2011). However, intrinsic factors, such as the age and genetic composition of the run, also influence abundance and the arrival timing of the composite run. Inclusion of age composition data available from test fisheries can improve in-season abundance estimates (Flynn and Hilborn 2004) and can assist in determining population-specific arrival timing and abundance in a mixed-stock fishery (Chasco et al. 2007), while the timing of the prior year's jack arrival can be used as a proxy for the effect of genetic composition on run timing (Anderson and Beer 2009).

Data on arrival timing of spring Chinook Salmon to the Columbia River were previously used in combination with a model of salmon migration based on geomagnetic cues to examine probable migration routes (Bracis and Anderson 2012). In the present study, we investigated whether age-specific arrival timing provided information on the oceanic distribution prior to homeward migration. Upper Columbia River basin spring Chinook Salmon (i.e., fish that spawn above Bonneville Dam) migrate to offshore regions and have an unknown oceanic distribution (Waples et al. 2004; Sharma and Quinn 2012); thus, mechanistic modeling of the migration routes provides a method for investigating hypothesized premigration distributions. Because a run comprises fish that have been in the ocean for 1–3 years or more (Healey 1991), the age-specific patterns in arrival within a season provide information with which to infer possible oceanic distributions of fish at the initiation of their homing migration. Specifically, we considered two hypotheses with regard to arrival timing: (1) arrival timing does not differ among populations; and (2) arrival timing does not differ by age. We also considered the following hypotheses about premigration oceanic distributions: (1) start location does not differ by age; and (2) start date does not differ by age or across populations. To evaluate these hypotheses, we extended our earlier analysis of the homing migration of salmon (Bracis and Anderson 2012) to demonstrate that (1) consistent with Keefer et al. (2004), different populations maintain consistent ordering of arrival despite interannual variation in arrival time; (2) within each population, arrival time is also ordered by age, with the oldest fish arriving first and the youngest fish arriving last; and (3) this pattern can be simulated with a geomagnetic homing mechanism that includes age-dependent swim speeds and a common start location and date (although jacks may also initiate migration later from an interannually consistent location).

METHODS

Data and Statistical Analysis

We determined adult arrival distributions of spring–summer Chinook Salmon that were PIT-tagged as juveniles in the Columbia River basin. The PIT Tag Information System (www.ptagis.org) provided detection dates of adult Chinook Salmon at Bonneville Dam (located 235 km upstream of the Columbia River mouth) in the years 2000–2011. Because fish were tagged as juveniles, it was possible to calculate age-specific (number of years in the ocean) mean arrival timing by year and population. To maximize the number of populations and years used in the analysis, we included cases with as few as 15 returning adults for a particular population, age, and year, although results were similar when we used larger minimum sample sizes. Summer-run upper Columbia River wild and hatchery populations, which are generally believed to be more genetically similar to fall Chinook Salmon than to spring Chinook Salmon (Myers et al. 1998; Waples et al. 2004), were excluded from analysis.

For hatchery fish, the hatchery of origin was used to group the fish (see Table A.1 in the online version of this article). Each of the included hatcheries had either a spring run or a summer run (except East Bank Hatchery, which had both runs, but only the spring run had sufficient returns for analysis). Lookingglass Hatchery was divided into Grande Ronde River and Imnaha River populations. Wild fish were classified into spring and summer runs that were distributed in six river regions (based on Beer 2008; see Table A.2 in the online version of this article). Although this is a broad classification and genetic distinctions likely occur on a smaller scale (Keefer et al. 2004; Waples et al. 2004), sample size considerations prevented us from using more specific groupings for examining arrival timing on population-specific, age-specific, and year-specific bases.

We compared the arrival timing of fish as a function of the number of years spent at sea (1-ocean [jacks], 2-ocean, and 3-ocean fish; 4-ocean fish were observed in wild populations and were combined with 3-ocean fish). All comparisons were performed on a population-specific and year-specific basis. We examined the ordering of populations across years by using the

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mean arrival time for each age (results were similar when using the median rather than the mean because although the arrival days were not necessarily normally distributed, the distributions were generally unimodal and symmetric). To determine the consistency of population ordering among years, we examined all pairs of populations across all years to see if the arrival order agreed or disagreed between years (or was missing), all for a given age. That is, "agreement" meant that for fish of a particular age-class, the mean arrival time of one population preceded the mean arrival time of another population in both years. We defined arrival order agreement (A) as the number of agreements divided by the number of comparisons (see appendix). A randomization test with 100,000 samples was performed to evaluate the significance of A for wild and hatchery fish spending 1, 2, and 3 years in the ocean; this method was used because we did not have data for all populations in all years necessary for standard tests of order agreement. The observed arrival order for each year was randomly shuffled, and the numbers of agreements and disagreements across pairs of populations and years were recalculated to compute A for each sample, which was then used to estimate the P-value (Ernst 2004).

In addition to examining the absolute ordering of populations, we also compared the relative timing of age-classes. We performed unequal-variance t-tests (Ruxton 2006) comparing the mean arrival times of 1-ocean and 3-ocean fish for each return year and population relative to the benchmark arrival time of 2-ocean fish (the most common life history strategy) from the same return year and population. This difference in means is represented by the symbol Δt. A positive Δt indicates that mean arrival occurs later than the corresponding mean arrival of 2-ocean fish for that population and return year; a negative Δt represents an earlier mean arrival timing relative to 2-ocean fish. The unequal-variance t-test was selected because of unequal sample sizes ($|n_1 - n_2|$, wild: mean = 108, range = 0–377; hatchery: mean = 235, range = 2–2,512) and unequal variances ($|SD_1 - SD_2|$, wild: mean = 4.5, range = 0.1–43.5; hatchery: mean = 2.9, range = 0.1–13.2), in which case the unequal-variance t-test outperforms the nonparametric Wilcoxon test (Stonehouse and Forrester 1998). The unequal-variance t-test is robust to violations of normality other than extreme skewness (Stonehouse and Forrester 1998), and nonnormal data are also not an issue for large n (Lumley et al. 2002); this test also has the advantage of interpretable confidence intervals. In addition, no correction for multiple comparisons was made, so we would expect around 5% of the 1-ocean and 3-ocean groups to significantly differ from the 2-ocean groups at the $\alpha = 0.05$ level if the null hypothesis of no difference in mean arrival time by age was true (Rothman 1990). A chi-square test was used to compare age (1-ocean and 3-ocean) and arrival order as indicated by the sign of Δt (positive or negative) for both wild and hatchery fish. We also calculated the phi coefficient, a measure of the association of the two variables. Phi ranges between -1 and 1, where -1 indicates complete negative association, 0 indicates no association, and 1 indicates complete positive association (however, the extreme values are only possible in a balanced table).

Simulation Model

We used a simulation model of oceanic salmon migration (Bracis and Anderson 2012) to explore the effects of migration start date, location, and swim speed on age-specific arrival timing. The model combines daily ocean currents from the Ocean Surface Current Simulations (OSCURS) model (Ingraham and Miyahara 1988, 1989) with fish behavior to compute a trajectory from a start location in the ocean to the river mouth. Given the lack of empirical data on estuary transit, we calculated an arrival date at Bonneville Dam (see Bracis and Anderson 2012: Appendix S1) by using the simplest assumptions of no delay in the estuary and no age-specific differences in behavior. The fish swimming direction is based on the geomagnetic imprinting hypothesis (Quinn 1982; Lohmann et al. 2008), which postulates that salmon imprint on a component of the geomagnetic field when entering the ocean as juveniles. During homeward migration, the cue difference (i.e., the difference between the imprinted and local values) guides the fish to the isoline matching the imprinted value and then along that isoline to the river mouth. Potential geomagnetic cues are inclination angle and intensity, the isolines of which run approximately east–west in the North Pacific, and the home isoline is the isoline matching the imprinted value when the fish returns (Figure 1). Two behaviors (approach and follow) describe how the fish adjusts its swimming angle according to the cue difference. When the fish is distant to the home isoline, corresponding to a large cue difference, the approach behavior directs the fish along a meridional heading toward the home isoline. As the fish nears the home isoline, the cue difference reaches a threshold and the fish switches to the follow behavior. At that point, the fish shifts its swimming angle closer to the home angle (the swimming angle at which the cue difference is zero) as the cue difference decreases, and thus the fish follows the home isoline to the river mouth. We applied the model to Columbia River spring Chinook Salmon (Bracis and Anderson 2012); from our prior study, the parameter set that resulted in the largest proportion of fish arriving at the river mouth in the appropriate time window (parameters: cue = inclination; cue response threshold = 1°; home angle = 125°) was selected for use in the present study. Other parameter sets produced qualitatively similar results with respect to the patterns of Δt with age.

We simulated arrival patterns by using the model and then compared them with the observed patterns in the PIT tag data for the years 2001−2011. To explore the effect of model parameters on the arrival patterns, a series of simulations varied (1) swim speed, (2) start location, and (3) start date.

Age-dependent swim speed.—We varied fish swim speed by age, since age is correlated with length and optimal speed is a function of length. Based on age and length studies conducted at Bonneville Dam (Kelsey and Fryer 2001, 2002, 2003; Miranda et al. 2004, 2005; Whiteaker and Fryer 2006, 2007, 2008;

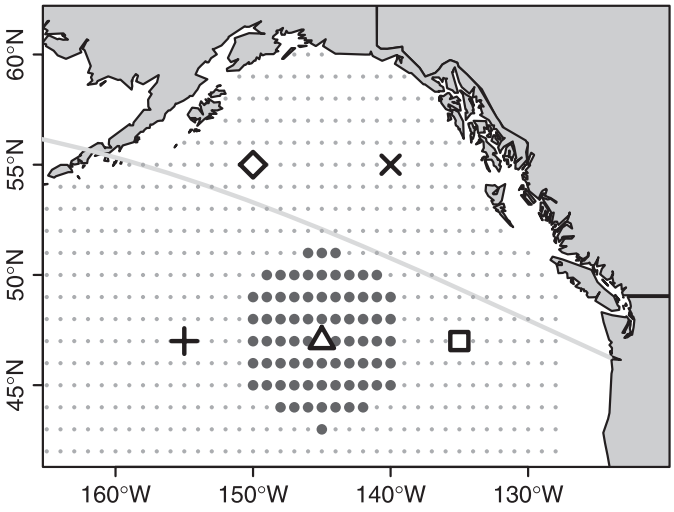


FIGURE 1. Start locations for Chinook Salmon in the simulation model. The five model start regions were centered on 55°N, 150°W (diamond); 55°N, 140°W (× symbol); 47°N, 155°W (plus symbol); 47°N, 145°W (triangle); and 47°N, 135°W (square); grid points are shown by small gray dots. Dark-gray circles illustrate the size of a representative start region centered at 47°N, 145°W. Light-gray line shows the home isoline for the inclination cue (see Methods).

Torbeck et al. 2009), the average length of spring Chinook Salmon is about 50 cm (SD = 4) for 1-ocean fish, 70 cm (SD = 4.5) for 2-ocean fish, and 85 cm (SD = 6) for 3-ocean fish. The model previously used a swim speed of 40 km/d, which is representative of the horizontal displacement over a day as demonstrated by previous modeling and empirical studies (e.g., Dat et al. 1995; Tanaka et al. 2005). We updated the model to use swim speeds of 30, 40, and 50 km/d for 1-, 2-, and 3-ocean fish, respectively; thus, we assumed that the smaller 1-ocean fish would swim slower than the 2-ocean fish and that the larger 3-ocean fish would swim faster than the 2-ocean fish. In additional simulations, we varied speed within age-classes by drawing swim speeds from a normal distribution with the SD (2.1, 2.6, and 3.5 km/d) based on the length SD. However, those results were very similar to the simulations without intra-age speed variability and are not presented here.

Start location.—To evaluate the possibility of a common start location producing the arrival time pattern of fish observed at Bonneville Dam, fish trajectories were simulated starting from points on a 1° × 1° grid in the eastern North Pacific bounded by 42°N, 165°W and the North American coastline (Figure 1). Each grid point represented about 8,000 km^2. We evaluated the effect of start location on arrival time in two ways. First, to characterize the effect of different large-scale starting regions, we used five starting regions consisting of 75 contiguous grid points around a central location (see Figure 1). The age-specific differences in mean simulated arrival times for each region (Δt) were calculated relative to the mean arrival time of 2-ocean fish in the same region. Second, to evaluate the effect of small-scale patchy distributions, we compared arrival times from all pairwise combinations of grid points. We characterized the spatial difference between patches in terms of the distance (Δx) between the selected grid points. For each pair of grid points, we calculated Δt values for two cases: (1) mixed-age patches, with fish of all ages started from both points; and (2) single-age patches, with fish of different ages started from each point. For example, in the first case, 1-ocean and 2-ocean fish were started from both points and their respective arrival times were averaged before calculating Δt. In the second case, the arrival time of a 2-ocean fish from one point was subtracted from the arrival time of the 1-ocean fish from the other point, and vice versa.

Start date.—To evaluate the possibility of age-dependent start dates producing the observed arrival time pattern, we also varied the start date. Our reference start date was March 21, the spring equinox. In addition to a single start date, we varied the migration initiation date by ± 10 d and ± 20 d around March 21. To investigate the possibility that fish of different ages initiate migration on different dates, perhaps due to some internal factor such as acquired energy stores or maturation threshold, we calculated Δt for 2-ocean fish starting March 21 and 1-ocean and 3-ocean fish starting earlier and later. In this case, we also examined 1-ocean and 3-ocean fish starting from different locations than 2-ocean fish. In addition, we simulated migration initiation dates ± 10 d and ± 20 d around April 20 under the assumption that late-arriving populations had later migration initiation dates than early arriving populations. The patterns of Δt for April 20 were very similar to those for March 21 but with later arrival times and are not presented here.

RESULTS

Observed Data

Order of arrival.—Both hatchery and wild populations arrived in a consistent order from year to year, with A equal to about 0.8–0.9 (Table 1). This was true for all age-classes whether considered separately or together. That is, if one population tended to arrive before another across years for 2-ocean fish, the same was also true for the 1-ocean and 3-ocean fish of those populations. The value of A was similar when only spring populations were included, so the known ordering of spring and summer runs did not drive the results (there were not enough summer-run populations to analyze separately). However, the comparisons were not spread evenly across populations and years, as there were more data in later years and for some populations (Table A.3 in the online version of this article). Within a population, the pattern of mean arrival time was similar among ages examined across years, with certain years tending to have earlier or later arrival times than other years; the same pattern also held across populations (Figure 2). For example, mean arrival was earlier in 2010 than in 2011 for all 2-ocean and 3-ocean fish of different populations and for 1-ocean fish of some populations. In general, the 1-ocean fish showed less interannual variability in arrival time than older fish.

Relative mean arrival by age.—There was a strong correspondence between age and arrival order: 3-ocean fish nearly

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TABLE 1. Values of arrival order agreement (A; the number of agreements divided by the number of comparisons) for wild and hatchery Chinook Salmon, where the number of agreements is the sum (over all pairs of populations) of the number of years in which a pair of populations arrived in the same order as calculated within age-classes and among all age-classes. The value of A was calculated for all populations and for spring populations only, since spring- and summer-run populations are demarcated by arrival time. The estimated P-value ($\hat{P}$) was obtained by performing a randomization test (Ernst 2004); n indicates the number of comparisons for which data were available.

		Wild			Hatchery		
Age-class	Run	A	n	$\hat{P}$	A	n	$\hat{P}$
1-ocean	Spring and summer	0.81	27	0.014	0.89	238	<0.001
	Spring only	0.83	12	0.059	0.88	160	<0.001
2-ocean	Spring and summer	0.90	468	<0.001	0.84	805	<0.001
	Spring only	0.83	241	<0.001	0.81	571	<0.001
3-ocean	Spring and summer	0.87	53	<0.001	0.79	39	0.006
	Spring only	0.76	21	<0.001	0.71	24	0.077
Across ages	Spring and summer	0.87	1,351	<0.001	0.85	3,013	<0.001
	Spring only	0.82	642	<0.001	0.82	2,080	<0.001

always arrived before 2-ocean fish, and 1-ocean fish nearly always arrived after 2-ocean fish (wild: $\chi^2 = 40.2$, $P < 0.001$; hatchery: $\chi^2 = 72.3$, $P < 0.001$; Table 2). The phi coefficient was 0.96 for wild populations and 0.92 for hatchery populations. Values of Δt were significantly different from zero in 36 of 44 cases for hatchery populations and in 69 of 85 cases for wild populations, whereas only 2 and 4 cases, respectively, were expected under the null hypothesis with no correction for multiple comparisons. The strong correspondence between age and arrival order remained after the removal of data for which Δt was not significantly different from zero (wild: $\chi^2 = 27.9$, $P < 0.001$; hatchery: $\chi^2 = 57.7$, $P < 0.001$).

Examination of Δt per population and return year for wild and hatchery populations showed that the magnitude of Δt varied across years for a particular population (Figure 3, top and middle panels; Table A.3). The magnitude of Δt was generally larger for 1-ocean fish than for 3-ocean fish, with an over threefold difference in mean Δt across wild populations (1-ocean: 20.8 d; 3-ocean: −6.2 d) and hatchery populations (1-ocean: 15.6 d; 3-ocean: −4.4 d). Similarly, for 1-ocean fish, 95% confidence intervals for Δt overlapped zero in only two instances, whereas for 3-ocean fish about half of the 95% confidence intervals for Δt overlapped zero. The greater number of confidence intervals overlapping zero for 3-ocean fish was due to more similar timing for 2-ocean and 3-ocean fish versus 1-ocean fish rather than to differences in confidence interval width. The average confidence interval width was actually larger for 1-ocean fish than for 3-ocean fish in wild populations (1-ocean: 18.0 d; 3-ocean: 11.2 d); the average width was the same for the two age-classes in hatchery populations (1-ocean: 9.3 d; 3-ocean: 9.2 d).

TABLE 2. Contingency table of mean arrival timing (d) for 1-ocean and 3-ocean Chinook Salmon relative to 2-ocean fish in wild and hatchery populations (see Figure 3). A perfect correspondence between age-class and relative arrival timing would result in zeroes for each cell in the diagonals (indicated by bold italic type).

	Wild		Hatchery	
Arrival relative to 2-ocean fish	1-ocean	3-ocean	1-ocean	3-ocean
Before	***0***	23	***0***	26
After	20	***1***	56	***3***

Simulation Model

Start location.—The simulation model, using age-dependent swim speeds and with all ages started from the same starting region, produced an arrival pattern similar to that observed, with the 3-ocean fish arriving first, the 2-ocean fish arriving second, and the 1-ocean fish arriving last (Figure 3, bottom panel). Fish that started closest to the river mouth had smaller magnitudes of Δt, while fish that started farther away or from more northerly regions had larger magnitudes of Δt. The modeled arrival times for each start region reproduced the sign and general magnitude of observed Δt. However, no single region matched the across-year variability in the magnitude of Δt that was observed for many populations.

In contrast to the above simulations, in which the start region was the same for all age-classes, we next explored the effect of patchy fish distributions. When starting fish from all pairwise combinations of grid points, the modeled Δt was within the observed range, irrespective of distance between release points, when all ages started from both points (i.e., mixed-age patches; Figure 4, gray lines). However, when fish of different ages started from different points (i.e., single-age patches), the modeled Δt only matched the observed Δt for half of the pairs of points once the distance between points reached approximately 500 km (Figure 4, black lines).

Start date.—When all ages started in the same region, small differences in start date among ages (i.e., 1-ocean and 3-ocean fish departed within 10 d of 2-ocean fish) still resulted in Δt

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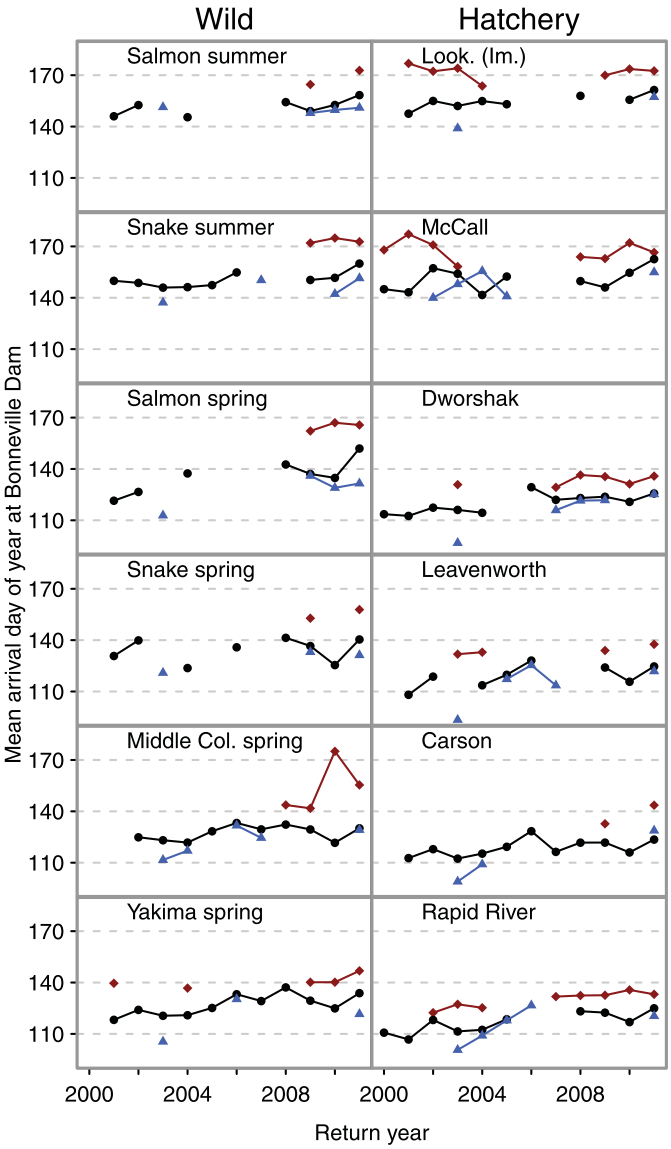


FIGURE 2. Mean arrival timing at Bonneville Dam for wild and hatchery Chinook Salmon populations by age across return years. Ages are 1-ocean (red diamonds), 2-ocean (black circles), and 3-ocean (blue triangles). Populations are identified by either river region or hatchery of origin (Look. = Lookingglass Hatchery; Im. = Imnaha River; Col. = Columbia River). Results shown are for the populations with the most data (results for all populations are available in Table A.3). [Figure available online in color.]

values within the observed range (Figure 5, diagonal). Next, we considered the case in which the start date and the start location differed between age-classes. For any particular start location of 2-ocean fish (Figure 5, columns), there was usually—but not always—a different starting location that could produce the observed Δt magnitudes for a particular start date offset. Additionally, 1-ocean fish that were located in the northeastern-most region (55°N, 140°W) and started 10–20 d after 2-ocean fish also produced the observed Δt magnitudes unless the 2-ocean fish started in the region closest to the river (47°N, 135°W).

DISCUSSION

Our results show that the ordering of mean arrival time to Bonneville Dam for PIT-tagged hatchery and wild Chinook Salmon populations was consistent across years and was in agreement with Keefer et al.'s (2004) observations of radio-tagged fish. Many studies have investigated the arrival timing of jacks (1-ocean fish) relative to older adult salmon (Jonsson et al. 1990; Shearer 1990; McKinnell et al. 1994; Morbey 2000; Carlson et al. 2004); however, our analysis based on tagged fish is the first to establish that age-dependent ordering of arrival occurs within individual populations of the Columbia River spring–summer Chinook Salmon run. Specifically, we determined that (1) within populations, the older (3-ocean) fish arrived first and the youngest (1-ocean) fish arrived last; and (2) within each age-class, the ordering of populations was also preserved across years. However, there were many missing values in our data, such that populations had only some years with arrival timing data for all three ages (Figure 2); thus, future data should be examined to verify whether this pattern holds. Interestingly, the variability in arrival time of the composite run across years (Keefer et al. 2008; Anderson and Beer 2009) was of the same magnitude ($\pm$ several weeks) as the differences in mean arrival between ages for the populations comprising the composite run. Part of the interannual variability of the composite run is the result of year-to-year variations in the fractions of early and late-returning populations making up the run (Anderson and Beer 2009), but another portion of the variability must involve differences in environmental conditions (Hodgson et al. 2006; Keefer et al. 2008). The fact that the composite run arrival date is variable but the populations within the run are ordered strongly suggests that environmental factors affecting fish return dates are acting on a regional scale. Here, we consider the preservation of arrival order in this highly variable system in terms of three interacting factors: the age-dependent swimming speed, the starting location, and the migration initiation date.

Simulations suggested that the apparent arrival ordering by age arose from age dependence in swim speed—meaning that fish spending longer in the ocean are larger and therefore swim faster. Simulated fish starting in the same region with age-dependent swim speeds replicated the pattern and magnitude of Δt observed in both hatchery and wild populations. This explanation is also compatible with the fact that different age-classes still showed a similar population ordering. Although for simplicity the simulations used a single start location for all age-classes, the same argument also holds if the initial distribution is patchy. As long as each patch comprises a mixture of all age-classes (i.e., rather than a scenario of separate age-classes starting in separate patches), the observed age-dependent arrival pattern emerges. It is possible that the pattern could arise from other mechanisms, but the simplest explanation for the consistency across years and populations is that the fish are intermixed when initiating homing, thus leading to the older, faster-swimming fish arriving first and the youngest arriving last.

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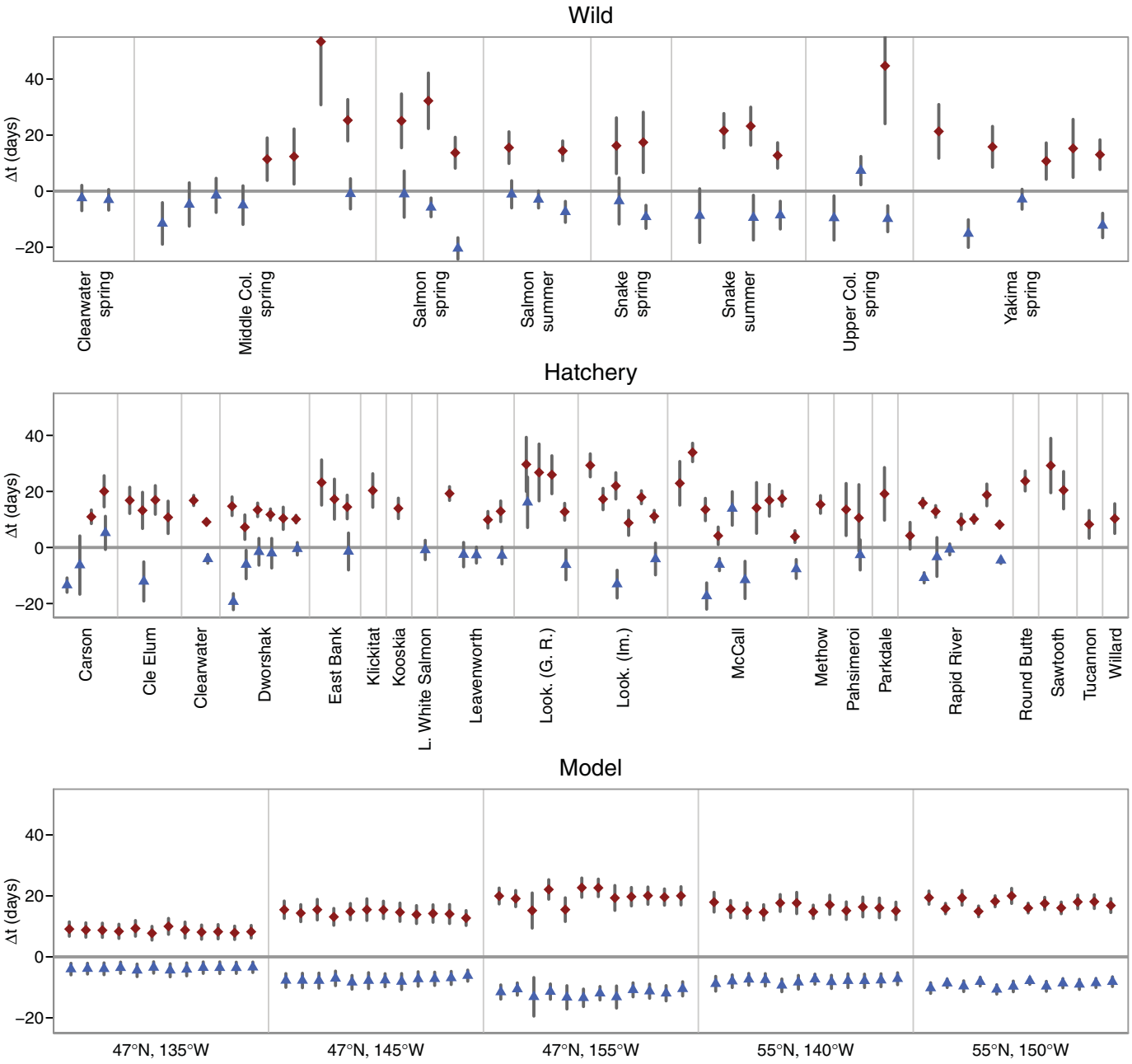


FIGURE 3. Population- and year-specific difference in mean arrival timing (Δt) of 1-ocean fish (red diamonds) and 3-ocean fish (blue triangles) in relation to 2-ocean fish (gray line at $\Delta t = 0$) for wild and hatchery Chinook Salmon populations and model results (Col. = Columbia River; L. = Little; Look. = Lookingglass Hatchery; G.R. = Grande Ronde River; Im. = Imnaha River). Points (ordered by return year) show Δt presented with 95% confidence intervals for t-tests comparing 1-ocean and 3-ocean fish with 2-ocean fish. Model results depict the effect of different start regions (see Figure 1) on Δt for consecutive years 2000–2011. [Figure available online in color.]

Since the oceanic distribution of Columbia River spring–summer Chinook Salmon is unknown, the mechanisms underlying this suggested commingling of ages (with the possible exception of jacks) at the start of their homeward migration are largely speculative. One possibility is that fish converge on a common location through tactic behavior involving prey availability (Francis et al. 1998; Mackas et al. 2007) or temperature preferences (Azumaya et al. 2007). Such biotic and abiotic conditions would be expected to vary interannually, which is also consistent with the interannual variation in the magnitude of Δt for individual populations, suggesting that the start region varies from year to year. Additionally, growth rate influences age at return (Parker and Larkin 1959; Neilson and Geen 1986; Quinn 2005); thus, perhaps fish that return in a given year are

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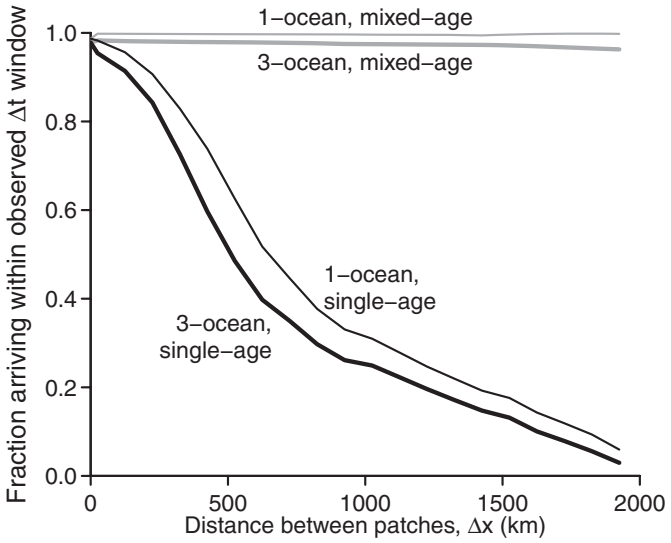


FIGURE 4. Effect of mixed-age patches versus single-age patches on the ability of model results to fit the observed range of differences in mean arrival timing (Δt) for Chinook Salmon: 0–30 d for 1-ocean fish, and −20 to 0 d for 3-ocean fish (relative to 2-ocean fish). Gray lines represent mixed-age patches (fish of all ages started from both points); black lines represent single-age patches (fish of different ages started from each point). Thin lines represent 1-ocean fish, and thick lines represent 3-ocean fish. Pairs of grid points (patches) are binned based on their distance apart (Δx) in 100-km increments. Nearly all simulated Δt values are within the observed range, regardless of distance between patches, when both patches contain all ages of fish. However, when patches do not contain all ages of fish, the fraction within the observed range declines with distance, falling below 50% once the patches are greater than 500 km apart. Results are shown for 2006 but were similar across years.

commingled based on the current or recent distribution of more productive patches.

However, the smaller interannual variability in arrival timing of jacks is surprising given their assumed slower swimming speed. Juvenile spring Chinook Salmon proceed rapidly northward along the coast toward Alaska (Trudel et al. 2009), but little is known about their subsequent ocean distribution, although it is believed to shift northward in summer and southward in winter (Myers et al. 2007). However, if jacks initiate their homeward migration at the same time as or after 2-ocean fish, it is possible that they are starting from an interannually consistent location in the northeastern Gulf of Alaska (region centered at 55°N, 140°W; Figure 5), while older fish vary more in their starting location across years, leading to larger interannual variation in their arrival timing.

We considered the hypothesis that all Columbia River populations shared a common migration initiation date, which implies that later-arriving fish would start farther away. However, the simulation results did not support this hypothesis. More distant start locations produced larger absolute Δt values, while the observed range of Δt magnitudes was similar between early and late-arriving populations (Figure 3; Table A.3). Thus, an alternative hypothesis—that migration initiation date is population specific—was supported by our simulations. Seasonal cues, such as photoperiod, are known to mediate hormone levels and reproductive state in migrating animals, which are important to migration timing along with acquired energy stores (Dingle 1996). Quinn (1982) proposed a population- and location-specific internal calendar based on circannual rhythms synchronized by photoperiod and adjusted for latitude. Hinch et al. (2006) noted that changing levels of reproductive hormones trigger migration and that initiation may also involve meeting growth or energy store thresholds since size affects the age at maturity. Although osmoregulatory capacity and sexual development appear to follow a fixed temporal trajectory in salmon (Crossin et al. 2009), energy stores and reproductive hormone levels also influence migratory timing (Crossin et al. 2007; Cooke et al. 2008). For Chinook Salmon, photoperiod explains much of the variation along a latitudinal gradient of spawning time in a gene that is central to the circadian clock of salmon, showing that photoperiod is a key cue in the timing of migration and that the response is heritable (O'Malley and Banks 2008; O'Malley et al. 2010).

If endogenous factors, such as energy stores, lead to variability in the migration initiation date, the effect may or may not vary with age. If the endogenous factors are similar across ages, resulting in only interannual variation in start date, then our hypothesis that all ages start in the same region is applicable. However, if age-classes respond differently (i.e., if migration initiation date is a function of age), then the commingling of ages is only feasible if the start date variation is less than $\pm$ 10 d across ages (Figure 5). Larger variations in age-dependent start date are possible only if different-aged fish start from different regions. However, the particular starting regions of 1-ocean and 3-ocean fish are still prescribed based on the initiation difference and the location of the 2-ocean fish, raising the question of why fish of different ages would be in those specific locations relative to one another. Age dependence in the start date also seems unlikely given the evidence for genetic control of spawning time (O'Malley et al. 2010).

Based on our analysis, we suggest that age-dependent arrival timing patterns may be useful for managing fish harvest. For example, because arrival is ordered by age, past data or preseason predictions of the age composition of runs could be used to improve run timing forecasts (e.g., Anderson and Beer 2009). Furthermore, while past studies have indicated that population arrival timing has a genetic component (Quinn et al. 2000; Waples et al. 2004), information on the age dependence of arrival timing could facilitate in-season management and the dispersal of harvest across populations and ages. For example, the arrival of 3-ocean fish would provide more precise predictions of the arrival of the 2-ocean and 1-ocean components of a run. Genetic stock identification (Hess et al. 2011), which is used to manage mixed-stock fisheries (Shaklee et al. 1999; Beacham et al. 2008), might be used together with associated age information (from tags or scale analysis) to establish in-season assessments of different populations' progress and vulnerability to the fishery. In turn, improved predictions of population-specific migration

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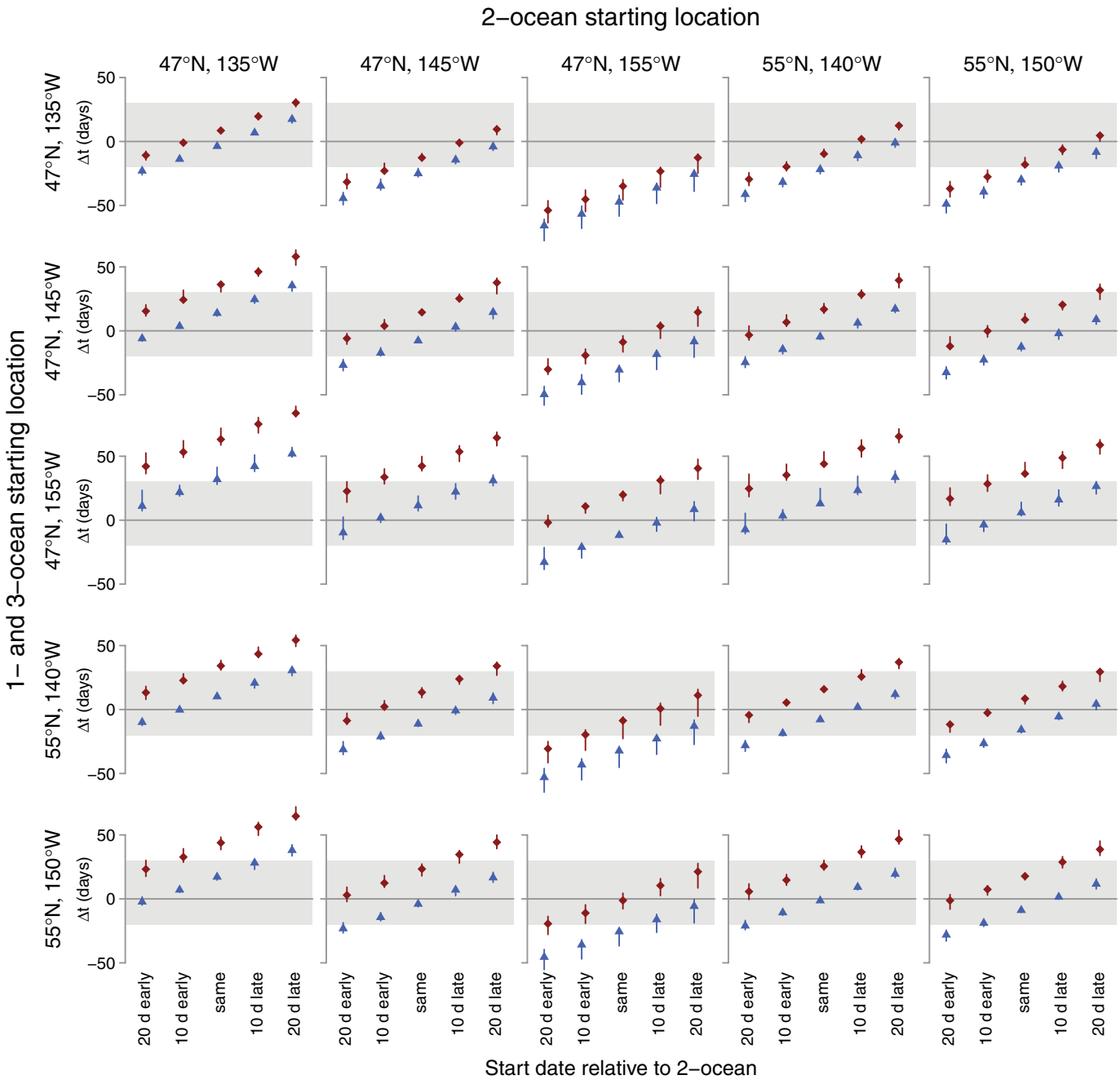


FIGURE 5. Difference in mean arrival timing (Δt) of modeled 1-ocean (red diamonds) and 3-ocean (blue triangles) Chinook Salmon relative to 2-ocean fish (gray line at $\Delta t = 0$), starting from all combinations of start locations with different migration initiation date lags. Rows show the start location of 1-ocean and 3-ocean fish, and columns show the start location of 2-ocean fish. Results across return years are plotted as box plots, with points showing the median and lines extending to the data extremes. Horizontal gray shading shows the approximate range of observed Δt values for 1-ocean fish (0–30 d) and 3-ocean fish (−20 to 0 d). Start dates for 1-ocean and 3-ocean fish are 10 or 20 d before, the same as, and 10 or 20 d after the start date for 2-ocean fish. [Figure available online in color.]

timing could be used to reduce the harvest vulnerability of endangered populations and to preserve the genetic diversity of the runs (Quinn et al. 2007).

In a broader context, our study suggests that patterns in arrival timing between and within distinct groups (e.g., genetic or age) can provide information on the migration strategies and oceanic distributions of salmonid populations. Our suggestion that different-aged fish in a population are intermixed (with the possible exception of jacks) stands in contrast to Weitkamp's (2010) finding that older fish tended to be recovered farther from their natal streams than younger fish. However, Weitkamp's (2010) analysis was based on coded wire tag recoveries in

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coastal marine fisheries, thereby excluding the upper Columbia River basin spring–summer Chinook Salmon, which have an offshore distribution and are thus not subject to coastal fisheries (Sharma and Quinn 2012). Generalizing this difference, if coastally oriented and offshore-oriented populations differ in how their oceanic distributions vary by age, then upper Fraser River spring Chinook Salmon, which also have an offshore distribution (Sharma and Quinn 2012), may also have a mixed-age oceanic distribution prior to their homing migration.

Differences in salmon oceanic distributions and migratory behaviors are also important to consider in relation to climate change (Chittenden et al. 2009). The summer habitat of salmon, as characterized by sea surface isotherms, is projected to shrink by over 40% by mid-century, with the Gulf of Alaska becoming winter-only habitat (Abdul-Aziz et al. 2011). This magnitude of change is likely to affect the homing start location of spring–summer Chinook Salmon and consequently would affect both the mean arrival timing and the age-dependent patterns in timing. The potential impact is unknown, in part because salmon migration timing is highly heritable and fish may adapt their homing initiation to compensate for changes in start location as well as changes in environmental conditions within freshwater habitats. A genetic-based shift in the run timing of Columbia River Sockeye Salmon *O. nerka* in response to warming river temperatures has been proposed (Crozier et al. 2011), and in Chinook Salmon the variability of a photoperiod-sensitive gene correlates with geographic patterns in the timing of spawning (O'Malley and Banks 2008; O'Malley et al. 2010). Weitkamp (2010) pointed to limited year-to-year variability in oceanic distribution to suggest that coastally migrating Chinook Salmon may be unable to adapt to climate change by shifting oceanic distributions. However, this conclusion may not apply to offshore-oriented populations, such as Fraser River and upper Columbia River basin spring Chinook Salmon, which exhibit more variable run timing than coastally oriented populations (Beer 2008) and thus are also likely to have a more variable oceanic distribution. From a larger perspective, the possibly different age-dependent distribution patterns in offshore- and coastal-migrating populations illustrate the complexity of salmon life history strategies. We suggest that further monitoring and analysis of age-dependent arrival patterns across salmon populations can contribute to understanding the driving ecological and evolutionary forces that shape those strategies and thus the potential responses of salmon to climate change.

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APPENDIX: CALCULATION OF ARRIVAL ORDER AGREEMENT

To develop the measure of arrival order agreement (A), we first calculated the mean arrival time for each age-class, population, and return year. For each age-class and return year, we converted the mean arrival values into values denoting the arrival order (U). That is, the first population to arrive (of a particular age-class in a given year) was assigned a U-value of 1, the second was assigned a U of 2, and so on. Thus, for each combination of age a, year y, and population p, we had $U_{a,p,y} = u$. Because we did not have comprehensive arrival information across populations and years for all ages, sometimes $U_{a,p,y}$ was missing (denoted "NA").

Next, we defined agreement for a pair of populations p and q in a pair of years y and z for a single age a as

$$V_{p,q,y,z} = \begin{cases} 1 & \text{if } \operatorname{sgn}(U_{a,p,y} - U_{a,q,y}) = \operatorname{sgn}(U_{a,p,z} - U_{a,q,z}) \\ 0 & \text{if } \operatorname{sgn}(U_{a,p,y} - U_{a,q,y}) \neq \operatorname{sgn}(U_{a,p,z} - U_{a,q,z}) \\ \text{NA} & \text{otherwise} \end{cases},$$

where sgn is the signum function that extracts the sign of the expression. In other words, V is equal to 1 if the populations arrived in the same order in both years; V is equal 0 if the order changed between years; and V is NA if any of the U-values are NA.

The A for a particular age-class was the sum of V over all pairs of populations and years divided by the number of comparisons,

$$A = \frac{\sum_{y=2000}^{2010} \sum_{z=y+1}^{2011} \sum_{p=1}^{n-1} \sum_{q=p+1}^{n} V_{p,q,y,z}}{N},$$

where there are n populations and where N is the number of comparisons (i.e., the number of times V is equal to 0 or 1 but not NA). In this case, we compared arrival order within an age-class (e.g., determining how often the order of mean arrival times of 2-ocean fish agreed between pairs of populations in different return years). To extend the comparison across ages, we summed over ages a and b,

$$A = \frac{\sum_{a=1}^{3} \sum_{b=1}^{3} \sum_{y=2000}^{2010} \sum_{z=y+1}^{2011} \sum_{p=1}^{n-1} \sum_{q=p+1}^{n} V_{a,b,p,q,y,z}}{N},$$

where V is defined as

$$V_{a,b,p,q,y,z} = \begin{cases} 1 & \text{if } \operatorname{sgn}(U_{a,p,y} - U_{a,q,y}) = \operatorname{sgn}(U_{b,p,z} - U_{b,q,z}) \\ 0 & \text{if } \operatorname{sgn}(U_{a,p,y} - U_{a,q,y}) \neq \operatorname{sgn}(U_{b,p,z} - U_{b,q,z}) \\ \text{NA} & \text{otherwise} \end{cases}$$

to additionally encompass agreement between a pair of ages a and b in addition to agreement between years.

To perform the randomization test, we took the $U_{a,p,y}$ values for each given age and year and randomly permuted them by assigning a random arrival order to each population. We then calculated A based on that random permutation. We repeated the randomization procedure $m = 100{,}000$ times, and we counted the number of times the sampled A was more extreme than our observed value, A^*. Thus, a P-value (Ernst 2004) could be conservatively estimated as

$$\hat{P} = \frac{1 + \sum_{i=1}^{m} I(A_i \geq A^*)}{m + 1},$$

where I is the indicator function.

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